

UNDERGROUND ECONOMY AND CORRUPTION

*Report
Submitted by*

PIYUSH SAWARKAR (22M0772)

CHIRANMOY BHATTACHARYA (22M0744)

SAYAN SURYA SHAW (22M0760)

As a Part of Course

**System Dynamics: Modeling & Simulation for
Development**

Under the Guidance of Our Instructor

PROF. OM DAMANI



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Indian Institute of Technology, Bombay

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Chapter 1

Motivation

The underground economy and corruption are two closely related phenomena that are pervasive in many societies around the world. The underground economy refers to all economic activities that are not reported to the government and are therefore not subject to taxation, regulation, or monitoring. This includes both legal and illegal activities, such as unregistered businesses, undeclared income, and black market transactions. Corruption, on the other hand, refers to the abuse of power or position for personal gain, often involving illegal or unethical practices such as bribery, embezzlement, and fraud.

The underground economy and corruption are intertwined because both involve the circumvention of legal and regulatory structures that are designed to prevent illicit activities and promote transparency and accountability. In many cases, corruption enables and perpetuates the underground economy by facilitating illegal transactions and shielding participants from detection and prosecution. Similarly, the underground economy can foster corruption by providing opportunities for officials to demand bribes or engage in other forms of illicit activity.

The consequences of the underground economy and corruption are numerous and far-reaching. They undermine the rule of law, erode public trust in government, and distort economic activity by favoring those who are willing and able to operate outside the legal framework. They also result in significant revenue losses for governments, which are unable to collect taxes on the unreported economic activity. Furthermore, they often exacerbate inequality and poverty by limiting opportunities for legitimate economic participation and perpetuating a cycle of illicit activity and corruption [OECD 2017](#).

Combatting the underground economy and corruption requires a multi-faceted approach that involves legal and regulatory reforms, enforcement of existing laws, and efforts to promote transparency and accountability. This may include measures such as improving tax collection systems, strengthening anti-corruption laws and institutions, increasing public awareness of the negative impacts of these practices, and fostering a culture of integrity and ethical behavior. Ultimately, addressing these issues is critical for promoting sustainable economic growth, reducing poverty, and building more

inclusive and just societies.

Chapter 2

Introduction

The system dynamics model outlined in this report pertains to the modeling of the underground economy, which encompasses both the black economy and the informal economy. While the term "black economy" typically refers to illegal economic activities that are intentionally concealed from society, the underground economy encompasses a broader range of activities that are not fully accounted for in the formal economy Medina and M. F. Schneider 2018. The informal economy, for instance, refers to economic activities that are not recognized by the government and are therefore not subject to regulation or taxation. The size of the informal economy is particularly significant in developing countries, although it constitutes a relatively small share of economic activity in developed nations. As such, in developed countries, the terms "black economy" and "underground economy" generally refer to the same illegal economic activities F. Schneider 2008.

From fig-2.1, we can notice the following things:

The underground economy can hinder a state's potential for growth and reduce its overall performance, leading to an increase in corruption and a decrease in government effectiveness, which can ultimately affect the quality of public infrastructure and impact the lives of citizens. Modeling the underground economy poses a challenge as its primary causes are not always easily identifiable or measurable. Furthermore, there is a dynamic inter-connectivity between the causes and effects, making it difficult to derive explicit cause-and-effect relationships. Buehn et al. F. Schneider and Enste 2013 highlights some of the critical causes and indicators of the underground economy. However, these causes and indicators are often interconnected. For instance, the GDP per capita and GDP growth rates can impact the economic stability of society, resulting in a reduction in corruption. Thus, these systems are highly interdependent, and modifying one variable can have a direct or indirect impact on the others Buehn and F. Schneider 2009.

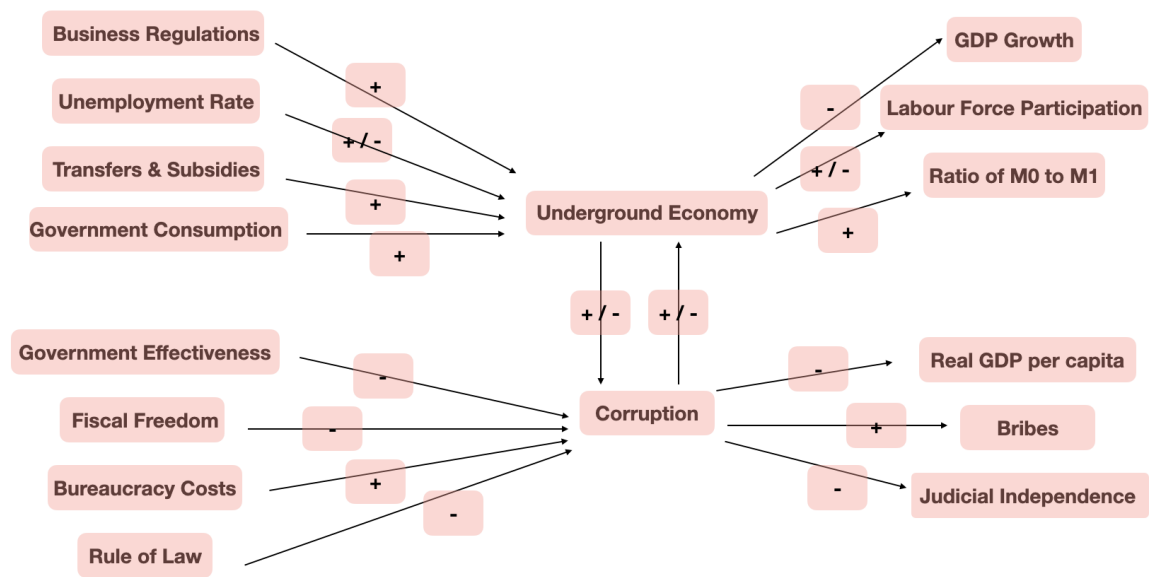


Figure 2.1: Dynamics of Underground Economy

Chapter 3

Model Details

We have divided the full model into 3 views. Explanation for each of the Views along with all the auxiliaries and the flows impacting the stock as well as the stocks is also given.

3.1 First View of the Model

The Vensim model presented in the given document represents a system dynamics simulation of a society's economic development in relation to public infrastructure spending, corruption, and other socio-economic factors. The model comprises various interconnected components that describe the effects of corruption acceptance, underground economy contraction, workforce migration, and other factors on the economy's growth rate. The model uses lookup tables to calculate additional GDP growth and corrupt-to-honest ratios and integrates differential equations to represent workforce migration and infrastructure degradation. The model's outputs provide insights into the impact of corruption and infrastructure investment on economic development over time. Please refer the [fig-3.1](#)

3.2 Second View of the Model

The Vensim model presented in the given document represents a system dynamics simulation of a society's economic development in relation to public infrastructure spending, corruption, and other socio-economic factors. The model comprises various interconnected components that describe the effects of corruption acceptance, underground economy contraction, workforce migration, and other factors on the economy's growth rate. The model uses lookup tables to calculate additional GDP growth and corrupt-to-honest ratios and integrates differential equations to represent workforce migration and infrastructure degradation. The model's outputs provide insights into the impact of corruption and infrastructure investment on economic development over time. Please refer the [fig-3.2](#)

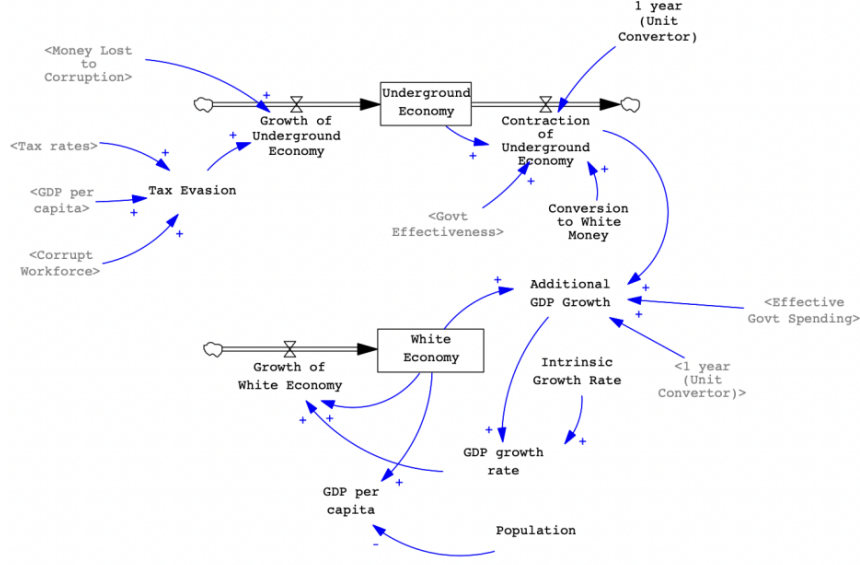


Figure 3.1: First View of the Model

3.3 Third View of the Model

The Vensim model presented in the given document represents a system dynamics simulation of a society's economic development in relation to public infrastructure spending, corruption, and other socio-economic factors. The model comprises various interconnected components that describe the effects of corruption acceptance, underground economy contraction, workforce migration, and other factors on the economy's growth rate. The model uses lookup tables to calculate additional GDP growth and corrupt-to-honest ratios and integrates differential equations to represent workforce migration and infrastructure degradation. The model's outputs provide insights into the impact of corruption and infrastructure investment on economic development over time. Please refer the fig-3.3

3.3.1 Explanation of all the Stocks and Flows Involved

The model Contains 5 Important Stocks. Explanation for each one of them is given below.

White Economy

The first stock "White Economy", represents the white GDP of a state i.e. The "White Economy" stock, which is the first stock, represents the economic potential available to a state in terms of its white GDP. The white GDP is treated as a stock with an inflow of "Growth of White Economy" and no outflow, as it is generally expected that

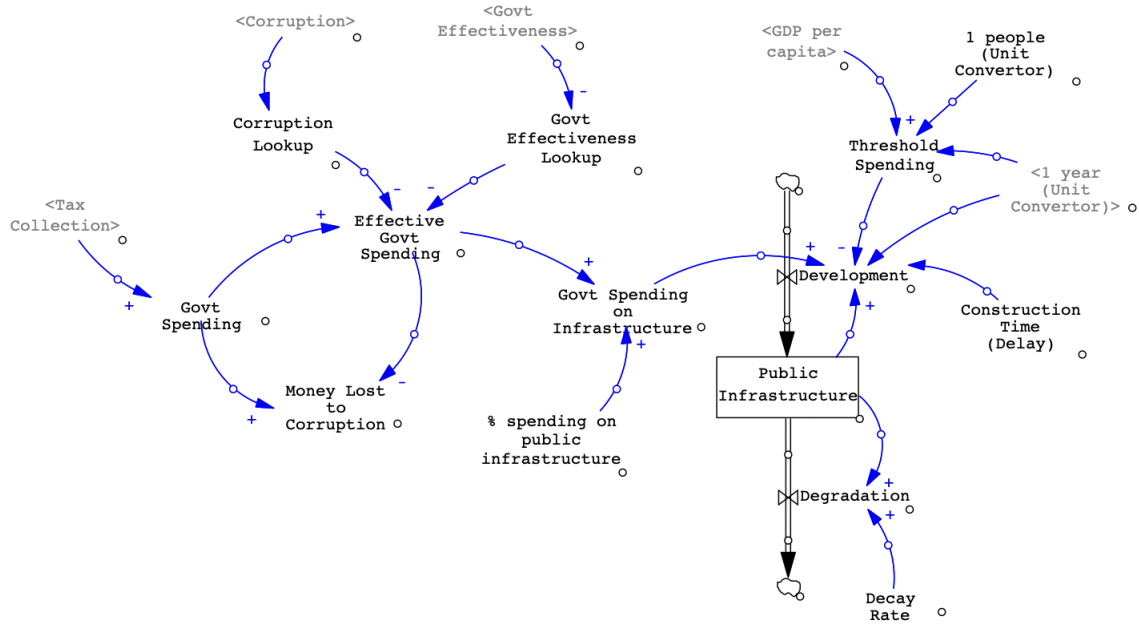


Figure 3.2: Second View of the Model

the GDP of a country will not decrease, except in rare circumstances such as a shock event like a pandemic, which is not included in this particular modeling. The growth of the white economy depends on two factors: the white economy itself and the GDP growth rate.

$$\text{Growth of White Economy} = \text{White Economy} * \text{GDP growth rate.}$$

Underground Economy

The "Underground Economy" is another component of this model and refers to the economic resources that are concealed from the state.

This model primarily encompasses two origins of illicit funds. The first is "Money Lost to Corruption," which signifies the funds that have been illicitly siphoned from government spending through corrupt practices, bribery, and other illegal means. The second source is "Tax Evasion," which refers to the income tax that is deliberately not paid to the government. The "Growth of Shadow Economy" is believed to rely on these two factors, as follows:

$$\text{Growth of Underground Economy} = \text{Money Lost to Corruption} + \text{Tax Evasion}$$

The underground economy experiences a decrease in its size due to a sink. This

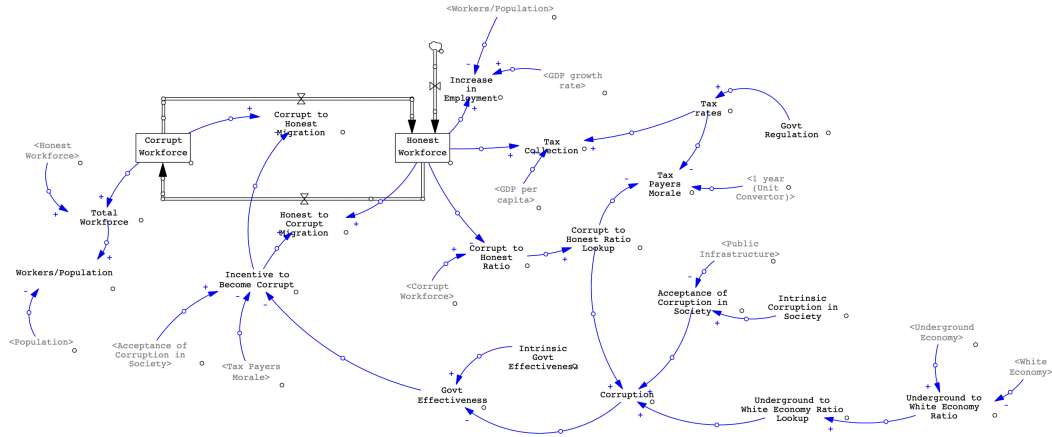


Figure 3.3: Third View of the Model

decrease is known as the "Contraction of the Underground Economy." It occurs when shadow money is either used to make purchases in the legal market, such as buying a house or a car, or when it is seized by legal means, such as through an income tax raid. The Contraction of the Underground Economy is determined by a combination of factors, including the level of spending in the legal market, the effectiveness of the government, and the size of the underground economy itself. To calculate the Contraction of the Underground Economy, the formula is:

(Spending in legal market + (Government Effectiveness/100)) * Underground Economy

Government Effectiveness is expressed as a number between 0 and 10, and to convert it to a rate, it is divided by 100. For example, a government effectiveness rate of 5 would lead to a 5% reduction in the shadow economy each year.

Public Infrastructure

The next stock is the "Public Infrastructure". It represents how the common Public goods benefit the citizens of a state, such as the quality of hospitals in a country. It is assumed that the quality of public infrastructure improves as the government spends more on it, relative to a threshold spending level that is linked to the GDP per capita of the country. This suggests that as the country becomes wealthier, more resources are required to construct the same infrastructure. Additionally, the rate of improvement in quality is also dependent on the current level of quality, as it takes more resources to improve something that is already of high quality.

To quantify the rate of growth in quality, a tanh function is used to calculate a value between 0 and 1 (for positive arguments) based on the government spending on public infrastructure divided by the threshold spending level. Dividing this value by 10 provides the growth rate, where a tanh value of 0.5 corresponds to a growth rate of 5%. However, it is important to note that the quality of infrastructure does not improve

immediately after the government spends money on it; there is a delay between the allocation of funds and the actual construction of the infrastructure. To account for this delay, the system incorporates a time lag.

Developement=DELAY3(Public Infrastructure*tanh(Govt Spending on Infrastructure/Threshold Spending)/10, "Construction Time (Delay)"/"1 year (Unit Convertor)"

The model assumes that the decline in quality of public infrastructure is affected by the overall quality of the infrastructure. Specifically, if the quality of the infrastructure is very high, then the rate of decay in quality will be greater. Therefore, the rate of decline in quality is not fixed, but instead changes with the level of quality of the infrastructure. This leads to a model in which the decrease in quality of public infrastructure is expressed as:

$$\text{Degradation} = \text{Decay Rate} * \text{Public Infrastructure}$$

Corruption

The following two stocks represent the number of citizens who earn money either through legal or illegal means.

The "Honest workers" stock represents the total number of civilians who pay their taxes, while the "Corrupt workers" stock represents the total number of citizens who evade taxes. There is assumed to be a flow between the two stocks, with individuals moving from one category to the other.

The growth rate of the number of workers is assumed to be dependent on the country's GDP growth rate, as it reflects the increase in the number of job opportunities. However, there is a limit to how much the number of workers can increase, as it cannot exceed the total population.

Therefore, it is assumed that the number of workers will keep increasing until it reaches a particular percentage of the total population. This leads to a model in which the "Increase in Employment" is expressed as:

$$\text{Increase in Employment} = \text{Honest Workers} * \text{GDP growth rate} / 4 * \text{"Workers/Population"}$$

The "Workers/Population" lookup provides a value between 0 and 1 based on the actual ratio of workers to population.

The flows of migration between honest and corrupt workers represent how the numbers of each group are changing within society. The direction of this migration, whether from honest to corrupt or from corrupt to honest, depends on the level of incentive to engage in corrupt practices. Specifically, when the incentive to engage in corrupt practices is high, there is a shift from honest to corrupt migration, while when the incentive is low, there is a shift from corrupt to honest migration.

$$\text{Honest to Corrupt Migration} = \text{Honest Workers} * \text{MAX}(0, \text{Incentive to indulge in corrupt practices}) / 100$$

Corrupt to Honest Migration = $-1 * \text{Corrupt Workers} * \text{MIN}(0, \text{Incentive to indulge in corrupt practices}) / 100$

The level of motivation for engaging in corrupt practices is determined by a combination of three variables, resulting in a score between -5 and 5 based on their values. These variables include the degree of societal acceptance of shadow work, the level of tax morale, and the effectiveness of government.

Incentive to Indulge in Corrupt practices = $(4 * \text{Acceptance of Shadow work in society} + (10 - \text{Tax payers Morale}) * 2 - 2 * \text{Govt Effectiveness}) / 6$

3.3.2 Explanation of all the Auxiliary variables and constants Involved

Tax Evasion

The term "Tax Evasion" refers to the amount of money that is hidden from the government in an illegal manner. Its value is determined based on the number of "Corrupt Workers," the tax rates, and the GDP per capita. It is assumed that the average income and GDP per capita of the country are similar, and as a result, tax evasion can be modeled as:

$$\text{Tax Evasion} = \text{GDP per capita} * \text{Tax Rates} * \text{Corrupt Workers} / 100$$

Additional GDP Growth

The variable "Additional GDP Growth" refers to the increase in GDP resulting from government spending and investment, as well as the eradication of underground economic activities. This variable has a direct impact on the actual GDP growth rate.

$$\text{Additional GDP Growth} = (\text{Effective Govt spending} + \text{Destruction of Underground Economy}) / \text{White Economy} * 1 \text{ year}$$

In order to ensure dimensional consistency, the variable "1 year" is used. The "Additional GDP Growth" variable increases in value as the ratio of government spending, the eradication of underground economic activities, and the white economy increase. The first two variables have a unit of "money per year," while GDP is measured in "money." Therefore, the "1 year" variable, with a unit of "year," is introduced to make the variable dimensionless.

GDP growth rate

It is assumed that the "GDP growth rate" is determined by both the "Additional GDP growth rate" and an "intrinsic growth rate."

$$GDP\ growth\ rate = (Additional\ GDP\ Growth/5 + Intrinsic\ Growth\ rate)/100$$

GDP per capita

The "GDP per Capita" is calculated as the total "White Economy" divided by the population of the country, which is assumed to be constant.

$$GDP\ per\ capita = White\ Economy/Population$$

Income Tax Collection

It is assumed that the "Income Tax Collection" is influenced by the total number of "Honest Workers," the "GDP per capita," and the "Tax rate."

$$Income\ Tax\ Collection = Honest\ Workers * Tax\ Rates * GDP\ per\ capita / 100$$

Tax Rate

The "Tax Rate" is adjusted based on the level of "Government Regulation," which is a value between 0 and 10 indicating the extent of the government's involvement in citizens' fiscal freedom. A higher level of government regulation leads to a higher tax rate.

Effective Government Spending

The modeled value of "Effective Government Spending" is obtained by subtracting the losses due to corruption and ineffective governance from the total amount of "Income Tax/Government Spending" collected. The impact of corruption and government effectiveness is adjusted through lookup variables that limit their effect within the range of 0-10.

$$Effective\ Government\ Spending = 0.1 * Govt\ Spending * (10 - (Corruption\ Lookup + Government\ Effectiveness\ Lookup) / 2)$$

Money Lost to Underground Economy

"Money lost to the Underground Economy" refers to the amount of money that is not captured in the official records due to its association with illegal activities. This is calculated by subtracting the Effective Government Spending from the Total Government Spending. The formula can be expressed as follows:"

$$Money\ Lost\ to\ Underground\ Economy = Total\ Government\ Spending - Effective\ Government\ Spending$$

Government Spending on Public Infrastructure

The amount of "Government Spending on Public Infrastructure" is calculated as a percentage of "Effective Government Spending" spent on infrastructure, where the assumed percentage spent on public infrastructure is 12

Mathematically, this can be expressed as: Government Spending on Public Infrastructure = Effective Govt spending * 12

Tax Morale

The variable "Tax Morale" represents the level of satisfaction that taxpayers have with regard to their contributions. It is influenced by both the tax rate and the ratio of corrupt workers to non-corrupt workers in society. Specifically, as the tax rate increases, tax morale tends to decrease, and as the ratio of corrupt to non-corrupt workers increases, tax morale tends to decrease as well. To quantify this relationship, the following formula is used:

$$\text{Tax Morale} = (10 - \text{Tax Rates} / (5 * "1 \text{ year}")) + 10 - (\text{Corrupt vs Honest ratio lookup}) / 2$$

Acceptance of Underground Work in Society

"Acceptance of Underground Work in Society" models how the general people in the society accept corruption. It decreases if the "quality of public infrastructure" increases. It also depends on the "Intrinsic corruption in Society" which is a measure of how corrupt a society is at the initial stage.

$$\text{Acceptance of Underground Work in Society} = (\text{Intrinsic Corruption in Society} + (10 - \text{Quality of Public Infrastructure}))$$

Corruption

The variable "Corruption" is used to measure the level of corruption in the society and is calculated as the average of three factors: "Acceptance of Underground work in society", "Corrupt vs. honest ratio lookup" multiplied by 0.7, and "Underground and White economy ratio lookup" multiplied by 1.3, all divided by 3.

Government Effectiveness

The measure of the government's ability to handle corruption in society is known as "Government Effectiveness." It is dependent on intrinsic government effectiveness, which is the effectiveness of the government in the absence of any corruption. However, in our model, government effectiveness is directly impacted by corruption. Still, corruption is not directly influenced by government effectiveness; rather, it indirectly depends on it. The formula for calculating government effectiveness is:

$$\textit{Government Effectiveness} = \textit{Intrinsic Govt Effectiveness} * (1 - \textit{Corruption})$$

Chapter 4

Impact of different variables on Final Outcome

This chapter studies the impact of different parameters on stock variables.

1. **Effect of Corruption and Effectiveness of Govt. on Infrastructure (construction time i.e. delay is 0):**

Expected Outcome: The level of corruption in infrastructure will be less severe if the government's initial effectiveness is high. Moreover, if the government is highly effective and corruption is low, then the development of infrastructure will be at its peak. Following this, the order of infrastructure growth will be high effectiveness and high corruption, low effectiveness and low corruption, and lastly, low effectiveness and high corruption.

Graph Generated [4.1](#)

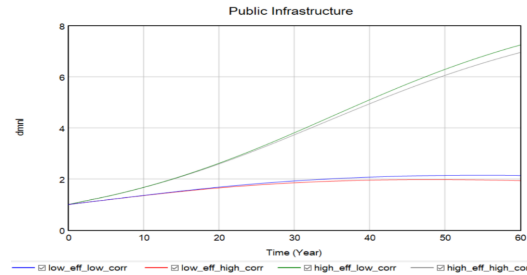


Figure 4.1

If the Government's Effectiveness is high, then the size of the underground economy must decrease. Conversely, if corruption increases, the size of the underground economy must also increase. Therefore, in a scenario where there is high government effectiveness and low corruption, the ratio between them should be at a minimum. Conversely, in a scenario with low government effectiveness and high corruption, the ratio should be at its highest.

2. Effect of Corruption and Effectiveness of Govt. on Infrastructure (delay=5):

Expected Outcome: When the delay in the system is prolonged, its impacts become more evenly distributed. Nonetheless, if the government's effectiveness is low, a delay in the system implies that the improvement in infrastructure is gradual. Thus, the influence of corruption and government effectiveness must have a greater impact on the quality.

Graph Generated 4.2

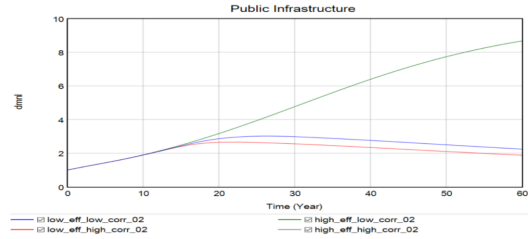


Figure 4.2

Our expectations were confirmed by the results, which indicate that corruption has a greater impact on government effectiveness when there is low effectiveness compared to the case with no delay. However, in the high-effectiveness scenario, the influence of corruption is almost indistinguishable from the no-delay case. This is because the delay period allows for five years of government effectiveness measures to counteract corruption, ultimately resulting in the two curves being identical.

3. Effect of Corruption and Effectiveness of Govt. on the ratio of Underground to White Economy (delay=0)

Expected Outcome: If the effectiveness of the government improves, it follows that the size of the underground economy will decrease. Conversely, if corruption increases, the size of the underground economy will also increase. Therefore, in situations where government effectiveness is high and corruption is low, the ratio between the two should be at its minimum, whereas in situations where government effectiveness is low and corruption is high, the ratio should be at its highest.

Graph Generated 4.3

Our expectations were confirmed by the results, which indicated that a lower level of government effectiveness corresponds to a higher ratio of underground economy to white economy, while a higher level of government effectiveness leads to a lower ratio.

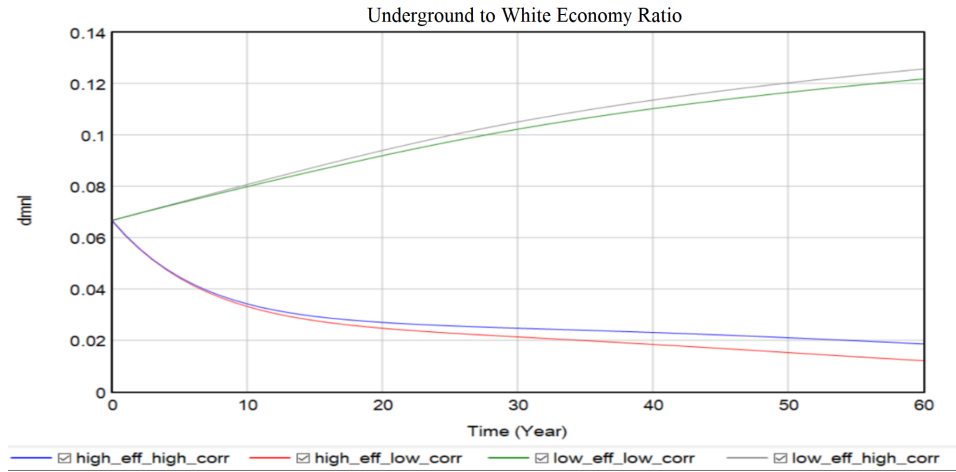


Figure 4.3

4. Effect of Corruption and Effectiveness of Govt. on the ratio of Underground to White Economy (delay=0)

Expected Outcome: As the delay lengthens, it is anticipated that the patterns will remain unchanged, but the impact of corruption is expected to intensify.

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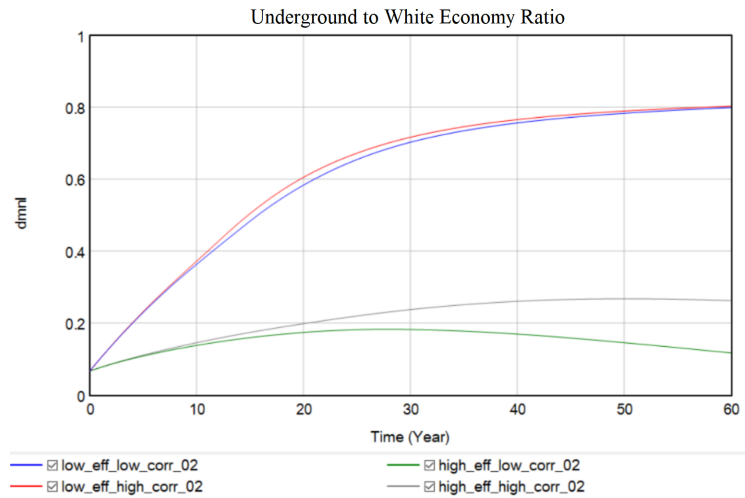


Figure 4.4

According to the findings, corruption has a greater impact on highly effective cases. Nevertheless, the proportion of the underground economy relative to the formal economy has risen across all cases.

5. Effect of Govt. Regulations on Infrastructure)

Expected Outcome: When the government increases its regulation on tax rates, it can have a negative impact on the tax morale of citizens, ultimately leading to an increase in underground work. While there may be a short-term increase in tax collection and subsequent growth in infrastructure, over time, corruption may hinder this growth.

Graph Generated 4.5

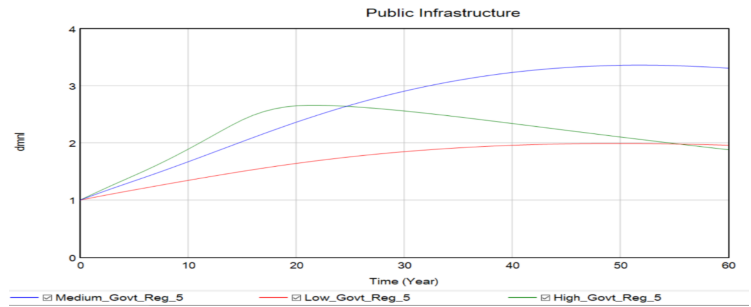


Figure 4.5

According to the findings, infrastructure experiences the most significant improvement when government regulations are high. However, this improvement quickly declines and falls below the level observed under low government regulations. This suggests that excessively high tax rates can have negative consequences. Nevertheless, when the tax rate is moderate, infrastructure continues to improve, indicating that there is a tax rate threshold that ensures sustainable growth. If the tax rate exceeds this threshold, it may result in a collapse later on, whereas a tax rate lower than the threshold may lead to suboptimal growth.

6. Effect of Govt. Regulations on the ratio of Underground Economy and White Economy)

Expected Outcome: With the increase in government regulations, taxes burden society more heavily, leading to a rise in the migration of individuals from the white market to the underground market. As a consequence, the underground economy experiences a rapid expansion, causing a proportional increase in the ratio of the underground to the white economy.

Graph Generated Figure 4.6

Based on the findings, it can be concluded that the growth of the underground economy is directly proportional to the increase in tax rates.

Graph Generated 4.7

It's important to acknowledge that high tax rates don't have a significant impact on the growth of the white economy. This is due to the fact that corruption increases dramatically in society, eroding the economy's potential to function.

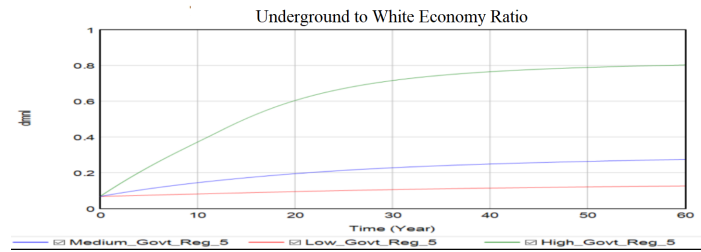


Figure 4.6

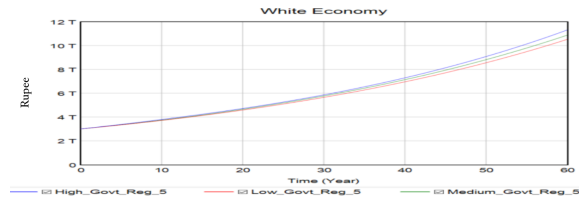


Figure 4.7

Although higher tax rates initially provide the government with greater resources that stimulate economic growth, this potential quickly diminishes.

Chapter 5

Data Source

The model is based on publicly available annual data from 2000-2005 of 51 countries, including 10 OECD countries, 6 non-OECD "high-income countries," and the rest are emerging markets. The countries can be categorized into two groups: industrialized countries with small shadow economies and low corruption levels, and developing/emerging countries with significant shadow economies and relatively high corruption levels. The data was provided by international organizations such as the World Bank.

Chapter 6

Challenges faced in Modeling

During the modeling process of the underground economy, several challenges were encountered. As the model was developed in different stages, each stage presented unique problems that had to be addressed before moving forward. Here, we share some of these challenges, in the hope that they may be helpful for other researchers looking to explore this topic.

1. Challenges Faced in Identifying Key Variables and Grouping Similar Variables:

In order to model any real-life system, it is important to identify the significant stocks and variables. During the modeling process of the underground economy, we encountered several variables that could be modeled as stocks, but adding them to the model would have increased its complexity without necessarily improving its effectiveness. Therefore, we needed to determine the most important variables that could be modeled as stocks.

For instance, in our initial attempt, we modeled corruption as a stock, but later we realized that it was challenging to model corruption using a stock. However, we recognized that there were other stocks that could model the effects of corruption in society. As a result, we modeled corruption as an auxiliary variable, which preserved its information without adding undue complexity to the model.

Identifying the significant auxiliary variables was another challenge we encountered in our modeling of the underground economy. Although various such variables have been reported in the literature, many of them are soft variables, and modeling them is not a straightforward task. Furthermore, some of these variables have similar effects on the economy, and modeling them all would add unnecessary complexity to the model.

For instance, government spending and government investments can be assumed to be two distinct variables, but they have the same effect on the economy, i.e., higher government spending or investment results in higher growth. Therefore, we

combined the effects of these two variables into one, without losing any relevant information.

2. Challenges in Determining the Impact of Variables on the Growth or Decline of a Stock:

During our modeling of the underground economy, we encountered several variables that had both positive and negative impacts on the growth or decline of a stock. This posed a challenge in deciding how to model these variables, as using the same variable to represent both growth and destruction could create difficulties in forming the equations.

For instance, consider the variable of corruption, which can lead to an increase in the number of corrupt workers. However, reducing corruption may also result in a decrease in corrupt workers. In this case, it is not sufficient to simply model the growth rate of corrupt workers, as the overall number of corrupt workers must also be taken into account. As a result, it can be challenging to properly identify the effect of corruption on the number of corrupt workers.

3. Challenges in identifying the Relation between Cause & effects:

In the process of modeling the underground economy, we encountered several variables that were highly coupled with one another, making it challenging to distinguish the causal variable from the affected variable. For instance, corruption and government effectiveness are two such variables that are interlinked in complex ways.

While the literature reports that corruption decreases as government effectiveness increases, it is also observed that as corruption decreases, governments become more effective due to more competent ministers and bureaucrats. Hence, it becomes difficult to determine their relationship. In our modeling approach, we assumed that corruption is a more general variable, and the less corrupt a society is, the better governance it exhibits. Therefore, we placed a direct arrow from corruption to government effectiveness. However, we also noted that government effectiveness has a strong but indirect impact on corruption, indicating a coupling between the two variables.

4. Challenges faced in Setting Up the equations with hyper-parameters:

Modeling the underground economy is a complex task, and the most challenging part is establishing the relationship between two variables. The scarcity of real-world data further complicates the matter. While one can have a qualitative idea of how a variable affects another, it becomes difficult to construct an equation that captures all the effects and their proportions when multiple variables influence a single variable.

The only viable approach to model such equations was to make an educated guess on the coefficients of each variable, simulate the entire model, and then update the

coefficients based on one's understanding of the outcomes. This iterative process allows for a better approximation of the relationship between the variables.

5. **Challenges faced in Modelling Soft Variables:**

Soft variables are those that are difficult to measure or quantify, such as social norms, beliefs, and attitudes. These variables are crucial in shaping the behavior of a system, and ignoring them can lead to inaccurate or incomplete models.

One of the main challenges in modeling soft variables is defining their values and relationships with other variables. Unlike hard variables such as GDP or population, soft variables are often dependent on individual perceptions and cultural factors. This subjectivity makes it challenging to define their values and quantify their effects on the system.

Another challenge in modeling soft variables is the lack of data. Since soft variables are difficult to measure, there is often limited or no data available to validate the model. To overcome these challenges, we have used qualitative data, opinions, and feedback loops to refine the model over time.

Chapter 7

Conclusion and Future Work

Measuring the underground economy with exact figures is a challenging task. In this study, several factors were considered to estimate the impact on different scenarios. It was revealed that the underground economy increases due to high corruption and lack of proper laws and regulations against bribery and illegal activities. However, the underground economy is not limited to illegal activities as it also includes the conversion of black or underground money into white money through illegal means, which was depicted in the stock flow conversion between white and underground economies. The growth of the underground economy is directly linked to an increase in corruption, corrupt workers, bribery, and strict policies against taxpayers, while the quality of public infrastructure and government effectiveness against these factors decreases. It is noteworthy that the money earned from the underground economy is sometimes spent directly in the formal economy. Therefore, a complete crackdown on the underground economy may not be the best approach. Instead, pushing toward the gradual transition of the underground economy into the formal economy could improve the white economy.

For future research, we can take national datasets and can incorporate additional factors specific to each country to obtain more accurate results. Though it is difficult to get all the economic growth and such type of underground economic data of a nation, we must try and compare it with the real trend.

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